

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. **Name of the Academic Unit: Mechanical Engineering**
2. **Subject Name: Mechanical Measurement and Mechatronics Laboratory**
L-T-P: 1-0-2 **Credits: 2**
3. **Pre-requisites: None**
4. **Syllabus and reference books:**

Syllabus:

Type of Signals associated with Sensors and Actuators

Signal Amplification and Conditioning (Inverting amplifiers and Instrumentation amplifiers)

DAQ systems to Acquire and Generate Signals

Computer Integration over Serial and Wireless Communication

Sensitivity, Resolution, Noise, Error, etc. associated with Acquired and Generated Signals

Measurement of Load, Angular Speed (RPM), Displacement, Velocity, Acceleration, Pressure, Temperature, Flow, Motor Current, etc.

Speed control of Motors (PWM DC Motor Speed control and Stepper Motor control)

Lecture will be delivered for one third of the classes and rest will practical.